

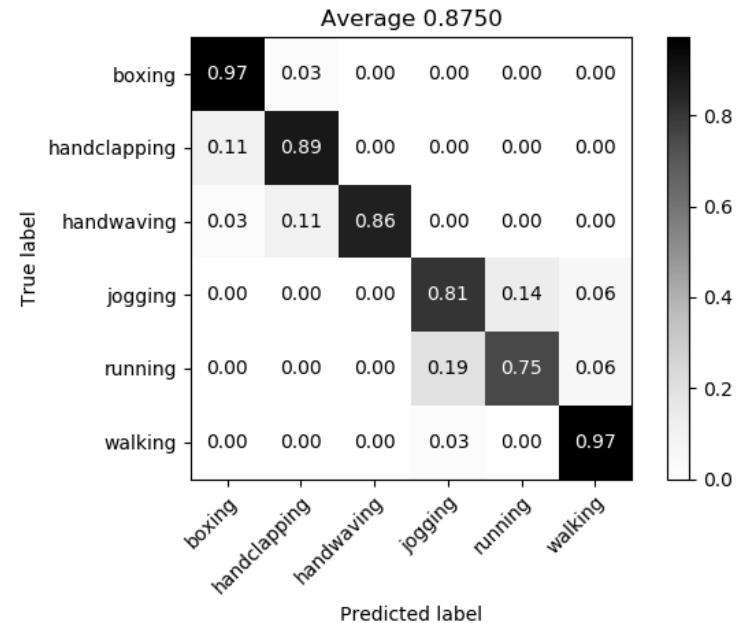
Exam Topics

Example Questions



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■ Example:



- ✓ ■ Boxing is never confused with handwaving
- Jogging is most difficult to classify



- Describe
- Sketch
- Multiple choice questions
 - Properties
 - Hyperparameters
- Recursive subdivision (Mosaic plot, treemap)
- Just noticeable difference
- Scagnostics
- Class consistency
- PCA
- Multi-dimensional scaling
- T-SNE
- UMAP
- [Hexagon] binning
- Kernel density estimation
- Multi-dimensional tables / data cubes
- Parallel aggregation
- Feature vectors (e.g., from neuron activations)
- Local perturbations
- Partial dependence
- Counterfactual explanation
- Force-directed graph drawing
- Layered graph drawing
- Graph morphing
- Foresighted graph layout
- ...



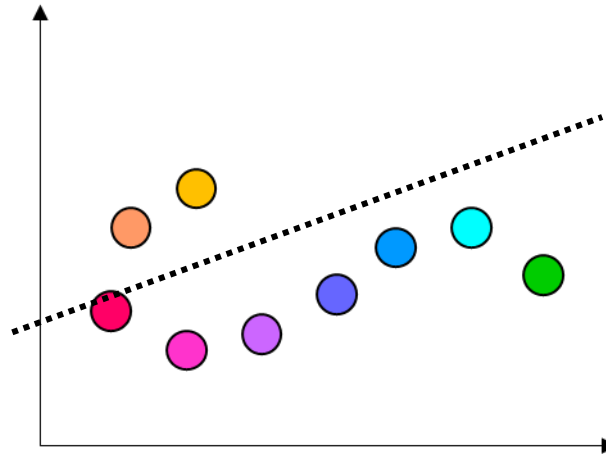
■ Example:

■ Select the correct statement(s):

- ✓ ■ Counterfactuals represent the minimal set of changes to feature values so that the prediction changes to a different output.
- Counterfactuals represent the maximum change of feature values so that the prediction switches to the opposite output.
- There is only one possible counterfactual explanation.
- ✓ ■ Counterfactuals should be chosen so that they are plausible.



- Example: sketch 1D PCA projection



- Given data and task, suggest and justify suitable
 - Data processing
 - Visualization design
 - Interaction design
 - Justify against alternative(s)
- Multivariate data
- Unstructured data
- Multidimensional data
- Tall data
- Large graphs
- Set-typed data
- Dynamic graphs
- ML model
- ...



- Given is a dataset about traffic accidents in Europe from 1970 to 2020. The **tabular data** has the following format:
 - Accident ID | longitude | latitude | year
 - The table has more than **10 million rows**. There are no missing data.
- How would you generate an **interactive visualization** (e.g., for the web) so that users can explore the distribution of accidents **spatially and temporally**?
 - Note: the visualization does not need to be multi-scale. Visualization on a coarse spatial level is sufficient.
- How would you **preprocess** the data?



- Preprocessing: we have a **scalability** challenge!
 - Binning into pre-defined regions and years → perceptual scalability
 - Data cube / multi-dimensional table → interactive scalability
- Visualization: two **linked** views
 - Spatial (regions can be brushed)
 - Choropleth map
 - (Hexagonal) bin map
 - Sequential color map
 - Temporal (individual years or year ranges can be brushed)
 - Line chart or histogram



- Point deductions:
 - Not identifying or handling of scalability challenge
 - No linked views → only spatial or temporal analysis possible
 - Ineffective processing (e.g., KDE)
 - Only spatial or temporal binning
 - Minor:
 - Missing names for visual encodings
 - Not mentioning binning or aggregation (just explaining it)
 - Not describing / naming interaction design
 - Not mentioning color mapping (or non-optimal color mapping)

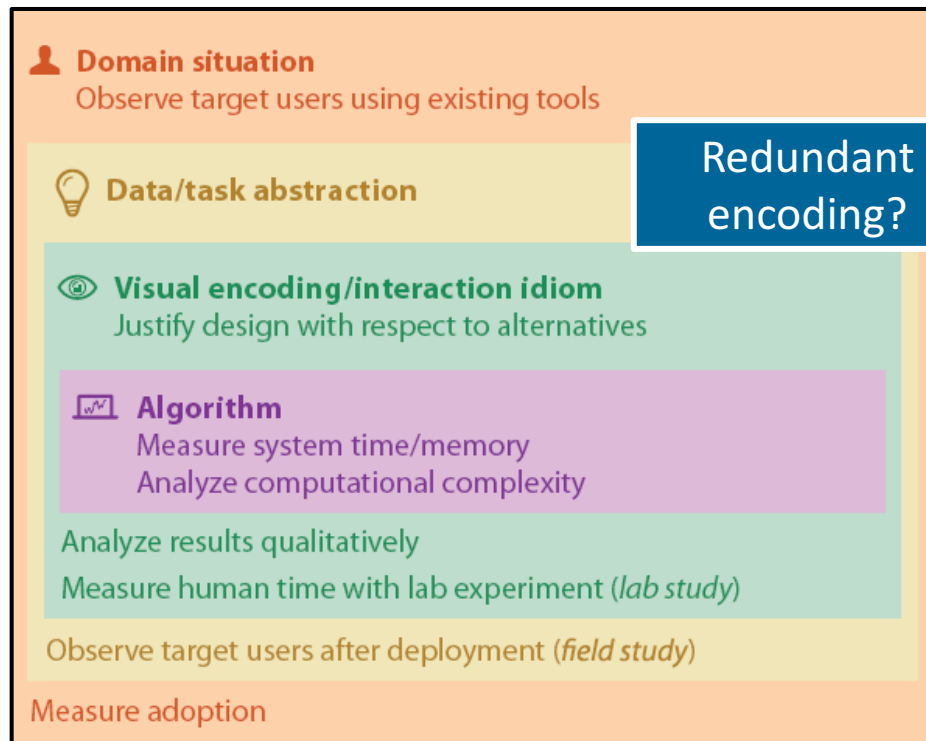


- Given a topic in visualization research, describe how you would validate your project

- Example (from an oral exam → too elaborate for written exam):
 - There are different options how to visually encode data items of different **classes (or sets) in static scatterplots**. How would you evaluate the effectiveness of possible visual encodings? Describe the design of your planned validation approach.



- Evaluation of visual encoding
 - Design alternatives



➞ Identity Channels: **Categorical** Attributes

Spatial region



Color hue



Motion



Shape



Spatial channel occupied

Categorical colors and shapes possible for static display

Marks as Links

➞ Containment



➞ Connection



Link marks possible for showing class association



- Hypotheses
 - Color will be more effective than shape [identity channel ranking]
 - Connectedness will be more effective than color [Gestalt Law]
 - Redundant encoding will be more effective than single-channel encoding [channel interference]
- Independent variable(s)
 - Visual encoding [color, shape, connection, color + shape, color + connection, shape + connection, color + shape + connection]
- Control / random / additional independent variables?
 - How many classes?
 - Balancing of classes?
 - How many points?
 - Distribution of points?
 - Chart size?

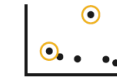


■ How to measure “effectiveness”?

■ Why do we visualize?

■ What are our low-level actions?

→ Identify



→ Compare



→ Summarize

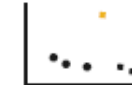


■ What are the data aspects of interest?

→ Trends



→ Outliers



→ Features



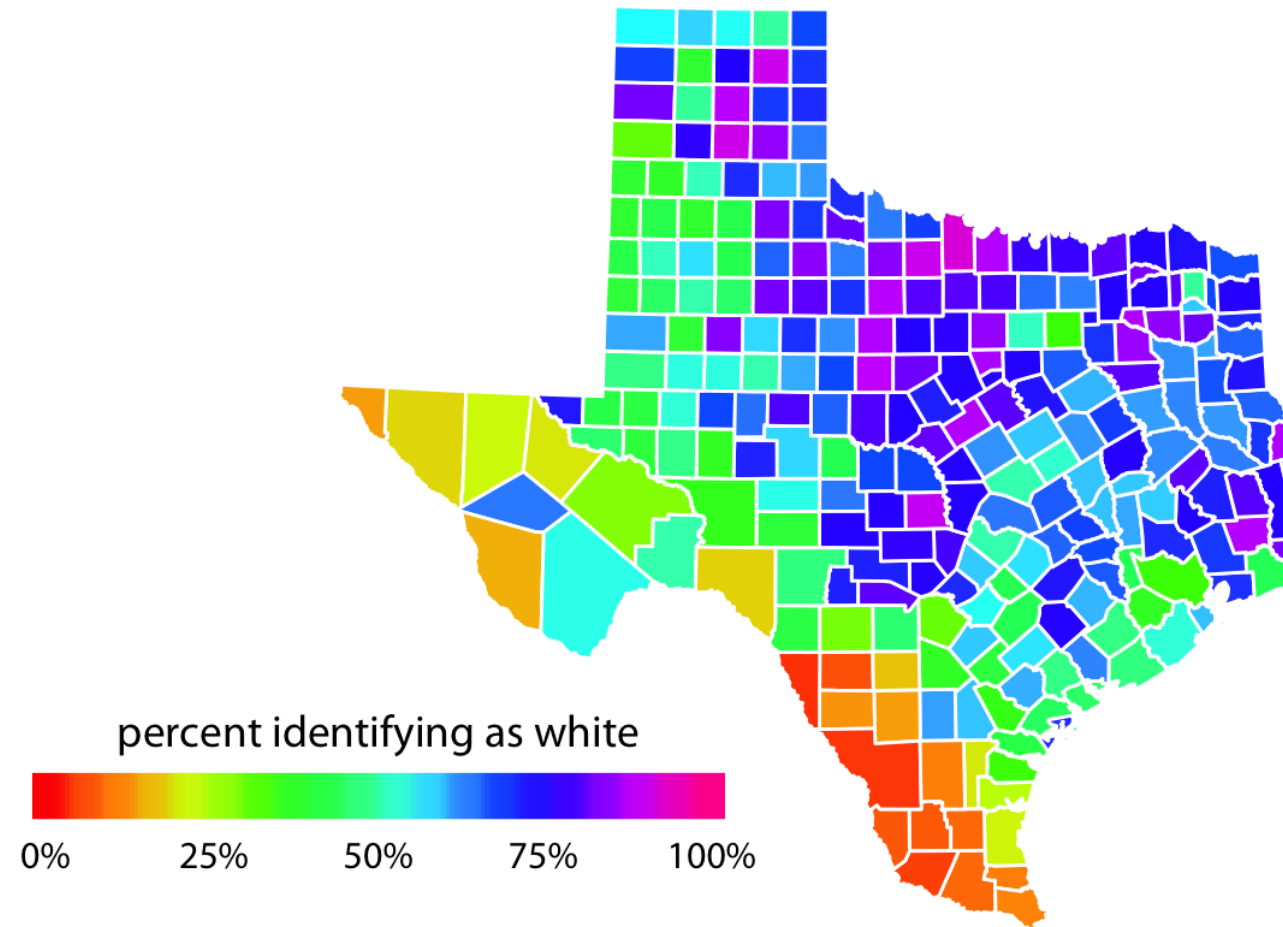
- How to measure “effectiveness”?
 - Task: click on all items that belong to a given class
 - Dependent variable(s)
 - Accuracy (false positives, false negatives)
 - Time?
 - Any confounding variables?
 - Crosses may be harder to select than circles



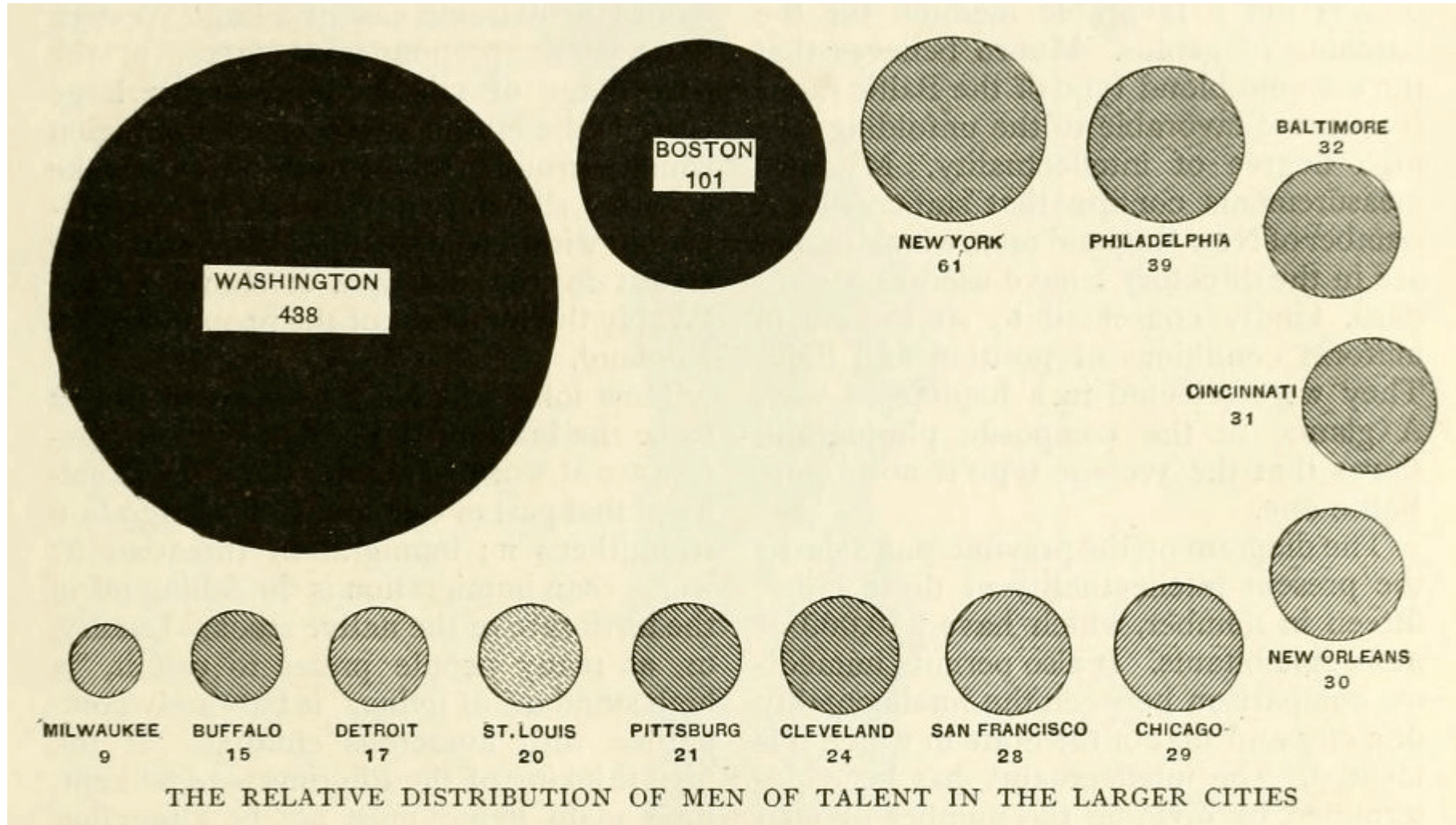
- Given a visualization:
 - Discuss strengths and potential problems, for instance based on
 - Data-ink ratio
 - Data aspects of interest and effectiveness
 - (Hidden) data transformations
 - Common design principles (e.g., 3D, “lie factor”)
 - Ordering (e.g., bars, axes)
 - Expressiveness & color maps
 - Effectiveness (e.g., Stevens’ power law)
 - Channel interference
 - Suggest alternative visual encodings



■ Example:

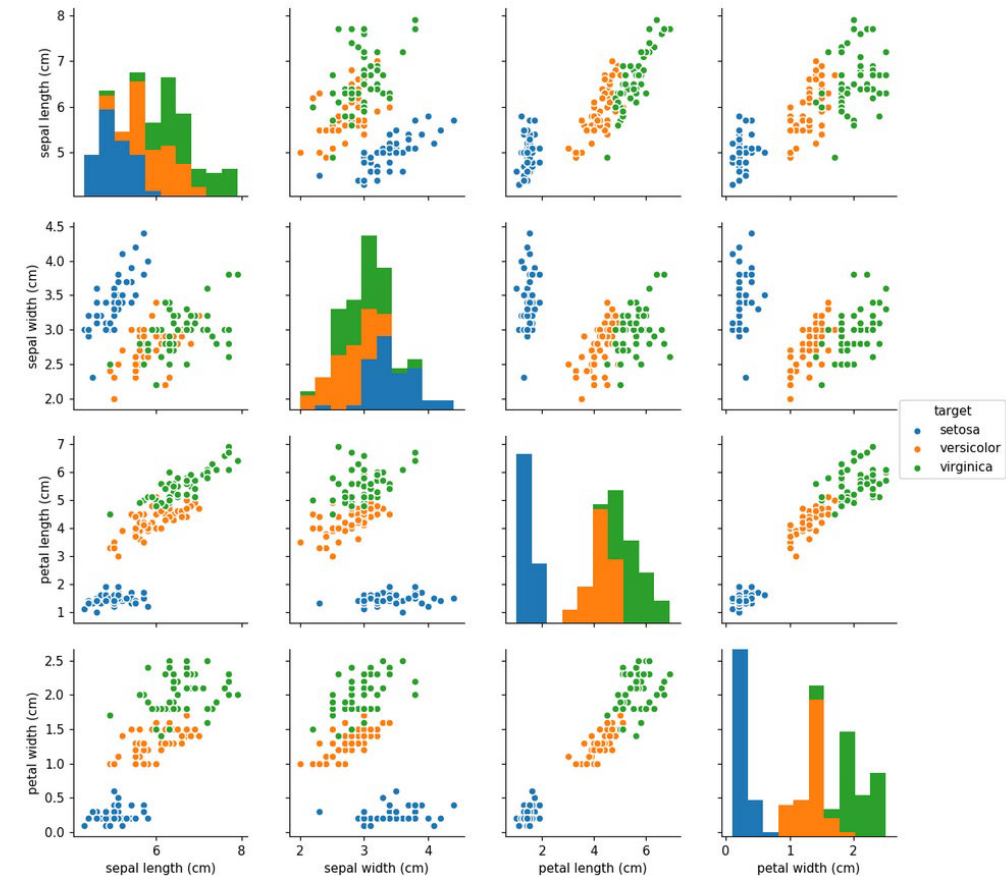
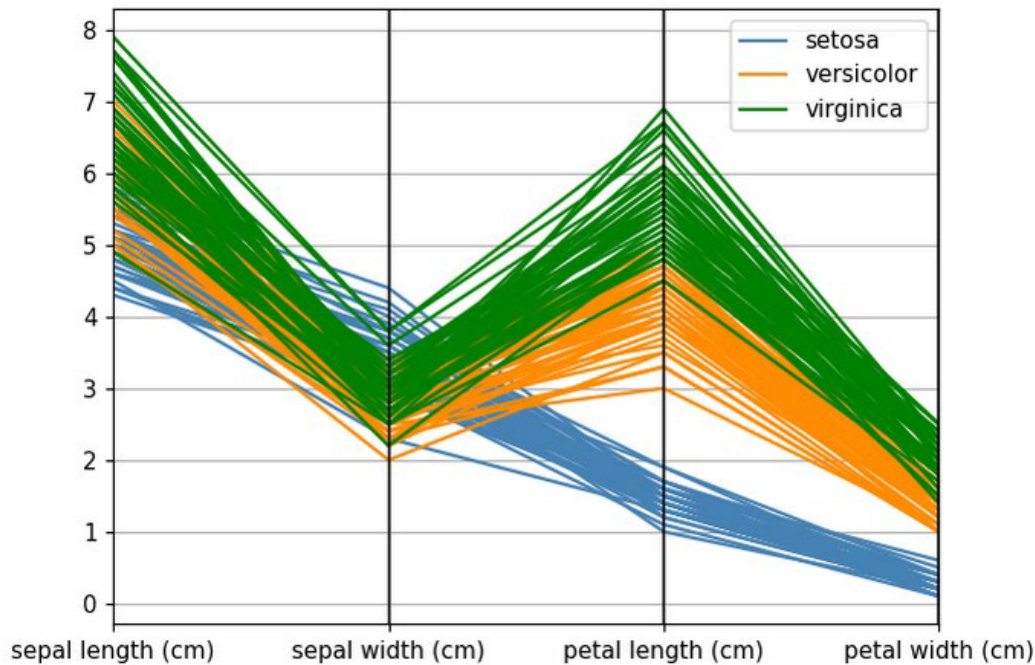


■ Example:



Alternatives

- Given two (or more) visualizations for the same data, discuss the advantages and disadvantages of the options
- Example:



- Transfer from one representation to another (sketch the solution)

- Example:
 - *Given:* [0, 5, 18, 22, 24, 34, 37, 38, 38, 39, 40, 41, 46, 49, 50, 58, 61, 66, 78, 85, 98]
 - *Sketch:* histogram, box plot



- Mark a data item highlighted in one visualization in the respective other visualization
- Examples:
 - Scatterplot $\leftrightarrow$ parallel coordinates
 - Parallel coordinates $\leftrightarrow$ star plot
 - Heatmap $\leftrightarrow$ grouped / stacked bar chart
 - Node-link diagram $\leftrightarrow$ adjacency matrix
 - Treemap $\leftrightarrow$ rectilinear node-link tree representation
 - ...



- List visualization-related concepts, for examples:
 - Raw data issues in data tables
 - Factors affecting scalability
 - Sources of latency
 - Aesthetics criteria for graph drawing
 - Simple solutions to overplotting that do not require any data preprocessing
 - Model-agnostic AI interpretability methods
 - Techniques to visualize set-typed data
 - Methods for validation of visual encoding and / or interaction design
 - Typical dependent variables in visualization user studies
 - ...

